From Zero Transfer to Structured Survival on Oriented-Circulant Baselines

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6 September 2026

Abstract

Building on established zero-transfer criteria for oriented circulants, we study a prescribed marked rank-one perturbation model that separates residual block darkness from survival under individually declared transitions. Standard finite-observability, invariant-kernel, polynomial-content and circulant Fourier methods are combined on fixed architecture and reachability strata. For a marked exact-dark pair, the standard rank-one resolvent update reduces the one-event model to two physical amplitude functions; in the associated one-dimensional marked realisation, the block and arrow content ideals are generated respectively by the feed-forward and return amplitudes. On the eight-site oriented-circulant zero-transfer baseline identified by Song and Lin, an affine family exhibits distinct block-dark and arrow-zero parameter loci. We also give a bounded seven-block calculation on the same baseline, reported as an exact worked classification without a priority claim. Static spectral data alone do not determine these parameter loci when the allowed amplitude functions vary. The paper therefore presents a reproducible structured-observability synthesis for prescribed perturbations of known zero-transfer baselines rather than a new general observability or cyclotomic theory.

Keywords: zero transfer; oriented circulants; structured observability; rank-one perturbation; parameter content; dark subspace.

1 Introduction

Zero transfer asks for pairs of vertices between which a continuous-time quantum walk has identically vanishing transition amplitude for all times. In finite-dimensional Hermitian models this condition admits equivalent spectral, resolvent and finite-moment formulations, and it has been studied directly in graph-theoretic settings by, among others, Sett et al. (2019) and Song and Lin (2026). For circulant and oriented-circulant baselines, Fourier diagonalisation makes the condition arithmetically explicit: spectral projector entries are finite sums of roots of unity, and vanishing becomes a statement about the corresponding masks and characters (Song and Lin 2026; Song 2026).

The present paper asks a deliberately narrower question. Suppose a source-target pair is already known to be exactly dark for an oriented-circulant baseline. What happens after one prescribes a parameterised perturbation, while retaining a marked decomposition into physically declared blocks and transitions? Two distinctions then become useful. First, an aggregate response may vanish through cancellation without any particular declared coordinate being individually dark. Secondly, a coordinate that is dark under the aggregate readout need not remain dark under every transition that the model declares separately. Our purpose is to organise established observability, invariant-subspace, coefficient-content and rank-one update tools around these two marked questions.

The mathematical mechanisms used below are standard or closely adjacent to established theory. Finite observability and Markov-parameter tests belong to classical systems theory (Kalman 1963); common invariant unobservable kernels occur in geometric control and switched-system realisation (Basile and Marro 1969; Petreczky et al. 2013); parameter-dependent polynomial certificates and systems over rings have a substantial literature (Habets 1992; Menini et al. 2020b; Menini et al. 2020a); coefficient content and principal generators are ordinary commutative algebra (The Stacks Project Authors 2026); prime-power root-of-unity cancellation is part of the classical cyclotomic literature (Lam and Leung 2000); and the scalar rank-one inverse update used in our one-event reduction is the Sherman–Morrison mechanism (Sherman and Morrison 1950; Hager 1989). We therefore do not present these ingredients as new general theory.

The contribution of this paper is deliberately bounded. We organise these established tools around a fixed marked perturbation model, attach parameter-content ideals to individual one-dimensional blocks and coherently recombined arrows, and use them to separate residual block darkness from survival under all declared continuations. We then work out a one-event rank-one family and a fixed seven-block eight-site calculation on a known zero-transfer circulant baseline. The seven-block calculation is retained as the principal bounded exact calculation of the paper and is reported without a priority claim. We do not claim the underlying observability, invariant-subspace, content, rank-one-update or cyclotomic mechanisms as new.

The scope is important. The baseline Hamiltonian is Hermitian and oriented-circulant, but the declared perturbation may be a directed rank-one coefficient. We do *not* claim that the perturbed operator remains an unweighted oriented-circulant adjacency, nor do we claim a universal taxonomy for arbitrary multi-event perturbations, higher-dimensional blocks, arbitrary composite orders, or genuinely multivariate parameter families.

2 Prior work and mathematical ingredients

2.1 Zero transfer and circulant arithmetic

Let H_0 be a finite Hermitian matrix and let s, ℓ be marked basis states. The scalar resolvent channel is

$$F_{\ell s}(z) = \langle \ell|(zI - H_0)^{-1}|s\rangle. \tag{1}$$

For finite dimension, $F_{\ell s} \equiv 0$ is equivalent to the vanishing of the transition amplitude $\langle \ell|e^{-\mathrm{i}tH_0}|s\rangle$ for all t, and to the finite family of moments $\langle \ell|H_0^k|s\rangle = 0$ through a Cayley–Hamilton horizon. These are standard spectral/observability facts; in the graph setting they form the basis of the zero-transfer literature (Sett et al. 2019; Coutinho and Godsil 2021; Song and Lin 2026).

For an n-vertex circulant, let

$$\omega = e^{2\pi\mathrm{i}/n}, \qquad |f_j\rangle = \frac{1}{\sqrt{n}}\sum_{r=0}^{n-1}\omega^{jr}|r\rangle.$$

If M_θ is the set of Fourier indices belonging to eigenvalue θ, the spectral projector is

$$P_\theta = \sum_{j\in M_\theta}|f_j\rangle\langle f_j|.$$

Define the marked Fourier sum

$$S_\theta(r) = \sum_{j\in M_\theta}\omega^{jr}. \tag{2}$$

Then

$$\langle u|P_\theta|v\rangle = \frac{1}{n} S_\theta(u-v). \tag{3}$$

Thus a marked pair is exactly dark precisely when the appropriate off-diagonal projector entries vanish on every spectral fibre. This is the circulant form used explicitly by Song and Lin (2026).

2.2 Finite observation and common invariant kernels

Consider a finite marked realisation with state space V, transition M and row observation Ω. Standard observability gives

$$\Omega(I-\eta M)^{-1}v \equiv 0 \quad \Longleftrightarrow \quad \Omega M^j v = 0 \quad (0 \le j < \dim V), \tag{4}$$

with the horizon replaceable by the observability index or minimal-polynomial degree. The same algebra underlies finite Markov-parameter tests in linear systems (Kalman 1963).

Our structured models additionally carry a decomposition

$$V = \bigoplus_{\gamma\in\Gamma} V_\gamma$$

and a finite set of declared arrows $L_\alpha : V_{s(\alpha)} \to V_{t(\alpha)}$. A subspace is called *decomposable* when it is a direct sum of block subspaces. If U denotes the blockwise unobservable part, the largest decomposable subspace of U invariant under every declared arrow can be obtained by the usual descending common-invariant-kernel construction. One may encode decomposability by adjoining the block projectors as letters and zero-extend each arrow to the full direct sum. This places the calculation within standard invariant/unobservable-kernel theory for switched or labelled systems (Basile and Marro 1969; Petreczky et al. 2013); quiver language provides another natural description of the block-and-arrow bookkeeping (Nijholt et al. 2020).

2.3 Parameter dependence and coefficient content

Systems over polynomial rings and algebraic certificates for parameter-dependent structural properties have long been studied (Habets 1992; Menini et al. 2020b; Menini et al. 2020a). In particular, Menini et al. (2020a) treat LTI systems whose matrices depend polynomially on a physical parameter vector β. They form maximal-minor ideals of PBH pencils in $\mathbb{R}[\beta, \lambda]$, use Gröbner covers to partition the physical parameter space into constructible branches, and give exact conditions for the parameter values at which the whole system is or is not observable. They also note that the classical Kalman observability test can be handled exactly by the same algebraic reasoning. Exact physical-parameter observability-loss loci are therefore established prior art.

Our target is narrower and coordinate-resolved. On a fixed marked architecture and reachability stratum, we ask whether one declared block has zero marked future identically in the spectral variable. After clearing only parameter-independent baseline denominators, the resulting finite numerator coefficients generate an ideal in the physical perturbation parameter t; on a one-parameter PID stratum its normalised generator is their greatest common divisor. This is an application of standard finite observability and coefficient algebra, not a new general observability criterion (The Stacks Project Authors 2026). What is specific to the present calculation is the marked block/arrow bookkeeping and its use in the prescribed zero-transfer perturbation model.

2.4 Prime-power root sums

Vanishing sums of roots of unity and their group-ring formulations are classical; see, for example, Lam and Leung (2000). For $n = p^a$, if a displacement r has character order $q = p^b > 1$, collapsing a mask modulo q reduces the vanishing test to divisibility by Φ_q. Since

$$\Phi_{p^b}(X) = 1 + X^{p^{b-1}} + \cdots + X^{(p-1)p^{b-1}},$$

the collapsed coefficient counts must agree across the p corresponding bands. We use this only as a convenient arithmetic test for local projector entries; it is not a classifier for the full parameterised response.

3 A marked perturbation model

Let H_0 be a fixed Hermitian baseline with spectral projectors P_θ, and let $A(t)$ be a prescribed parameterised perturbation. We work on a strong architecture/reachability stratum

$$\Sigma = V(I_\Sigma) \cap D(S), \qquad A_\Sigma = S^{-1}(R/I_\Sigma), \qquad R = K[t], \tag{5}$$

where the response coefficient field K contains the cyclotomic Song field $K_{\text{Song}} = \mathbb{Q}(\zeta_n)$, the fixed spectral poles and the fixed amplitudes required by the model. The point of (5) is to separate parameter values at which the declared architecture itself changes from parameter values at which a response coefficient happens to vanish.

We suppose the chosen realisation has one-dimensional declared blocks

$$\mathcal{R} = \bigoplus_{\gamma \in \Gamma} R_\gamma, \qquad \dim R_\gamma = 1,$$

and a finite set of coherently recombined declared arrows. The model is fixed on Σ; when coefficient-image ranks collapse, the physical realisation is rebuilt and absent coordinates are not interpreted as new dark states.

Definition 3.1 (Residual block darkness)**.** For a declared block R_γ, let $r_{h,\gamma}(z,t)$ denote its exact marked residual at grade h through a finite safe observability horizon H. The block is *residually dark* at a parameter value $t = \tau$ when all these residuals vanish after specialisation.

Let $K^{(0)}(\tau)$ be the direct sum of the reachable one-dimensional blocks that are residually dark at τ. Let $G_{\text{nz}}(\tau)$ be the directed graph of declared arrows whose specialised coefficients are nonzero.

Definition 3.2 (Structured survival)**.** $K_{\max}(\tau)$ is the largest decomposable subspace of $K^{(0)}(\tau)$ that is invariant under every specialised declared arrow.

For one-dimensional blocks the invariant-kernel computation has a particularly transparent graph form.

Proposition 3.3 (One-dimensional survival)**.** *Let $Z(\tau)$ be the set of reachable blocks contained in $K^{(0)}(\tau)$. Then*

$$Z_\infty(\tau) = \{\gamma \in Z(\tau) : \text{Reach}_{G_{\text{nz}}(\tau)}(\gamma) \subseteq Z(\tau)\}$$

and

$$K_{\max}(\tau) = \bigoplus_{\gamma \in Z_\infty(\tau)} R_\gamma.$$

Proof. After adding the block projectors and zero-extended arrows to the transition alphabet, standard common-invariant-kernel theory applies. In one dimension a decomposable invariant subspace is determined by a set of block vertices. Such a set is invariant exactly when every nonzero declared arrow issuing from it lands inside the same set. Hence the greatest invariant subset of $Z(\tau)$ is precisely the forward-closed set displayed above. This is the coordinate form of the usual greatest common-invariant unobservable kernel (Petreczky et al. 2013). □

The distinction between $K^{(0)}$ and $K_{\max}$ is therefore not a new invariant-subspace theorem; it is a modelling distinction. Aggregate darkness answers whether a marked coordinate is invisible to the output. Structured survival asks whether that invisibility is preserved under every transition the model insists on treating separately.

4 Block and arrow parameter contents

Fix a reachable one-dimensional block R_γ. Assume that the finite residuals are proper rational functions of z and that a parameter-independent monic polynomial $D(z) \in K[z]$ clears all fixed spectral denominators through the chosen safe horizon H. Write

$$D(z)r_{h,\gamma}(z,t) = P_{h,\gamma}(z,t) = \sum_j c_{h,j}(t)z^j. \tag{6}$$

Definition 4.1 (Block content)**.** The block-content ideal on Σ is

$$J_{\gamma,\Sigma} = (c_{h,j} : 0 \le h \le H)A_\Sigma. \tag{7}$$

When A_Σ is a PID domain, define

$$g_{\gamma,\Sigma} = \begin{cases} 0, & J_{\gamma,\Sigma} = (0), \\ \text{the chosen normalised generator of } J_{\gamma,\Sigma}, & J_{\gamma,\Sigma} \neq (0). \end{cases}$$

Thus $g_\gamma = 0$ denotes an identically dark block, $g_\gamma = 1$ a block that is never dark on the stratum, and a nonconstant generator a proper algebraic darkness locus.

Proposition 4.2 (Block content in the prescribed model)**.** *Under the fixed-denominator and finite-observability hypotheses above, a reachable block is residually dark at a geometric parameter point $\tau \in \Sigma$ if and only if*

$$J_{\gamma,\Sigma} \subseteq \mathfrak{m}_\tau.$$

On a one-parameter PID stratum this is equivalent to $g_{\gamma,\Sigma}(\tau) = 0$.

Proof. By finite observability, the entire future vanishes exactly when the finite list through H vanishes. The parameter-independent denominator does not create or remove a zero after specialisation, so this is equivalent to the vanishing of every polynomial $P_{h,\gamma}(z,\tau)$. Uniqueness of polynomial coefficients gives $c_{h,j}(\tau) = 0$ for every h, j, which is equivalent to containment of the generated coefficient ideal in the specialisation maximal ideal. On a PID stratum the same ideal is generated by g_γ. The use of a coefficient ideal, and its reduction to a gcd in a PID, is standard algebra (The Stacks Project Authors 2026); the proposition simply applies it to one marked observability column. □

The construction is independent of any fixed-pole coordinate basis: an invertible K-linear change of rational-function coordinates changes the coefficient vector by a matrix whose determinant is a unit in A_Σ, and therefore preserves the generated ideal. A parameter-dependent denominator is not admissible, because it can manufacture false parameter factors.

For a declared one-dimensional arrow e, coherently add all pieces belonging to that same declared event before testing whether the arrow vanishes. Write

$$\lambda_e(z,t) = \frac{Q_e(z,t)}{\Delta_e(z)}, \qquad Q_e(z,t) = \sum_k \eta_{e,k}(t) z^k,$$

where the fixed denominator Δ_e is parameter-independent on the stratum.

Definition 4.3 (Arrow content)**.** Set

$$E_{e,\Sigma} = (\eta_{e,k}) A_\Sigma.$$

On a PID stratum write its normalised generator as $h_{e,\Sigma}$, with the same zero/unit convention as for g_γ.

Proposition 4.4 (Arrow disappearance)**.** *At a parameter point $\tau \in \Sigma$ the declared arrow e vanishes identically as a rational function of z if and only if*

$$E_{e,\Sigma} \subseteq \mathfrak{m}_\tau.$$

On a PID stratum this is equivalent to $h_{e,\Sigma}(\tau) = 0$.

Proof. The specialised rational function vanishes if and only if its numerator polynomial does, and the latter holds if and only if every coefficient vanishes. This is elementary coefficient algebra; the substantive modelling requirement is that coherent recombination be carried out before forming the ideal. □

Combining propositions 3.3, 4.2 and 4.4 gives the practical terminal description used below:

$$\big(\Sigma_{\text{reach}}, \{g_\gamma\}, \{h_e\}\big) \quad \Longrightarrow \quad K^{(0)}(\tau),\ G_{\text{nz}}(\tau),\ K_{\max}(\tau). \tag{8}$$

Equation (8) is a convenient organisation of standard finite observation, coefficient content and common invariant kernels; it is not presented as a new general observability theorem.

Remark 4.5 (Common-future divisor)*.* For a nonzero residual list on a PID stratum, g_γ is nonconstant precisely when every coefficient $c_{h,j}$ shares a nonunit common divisor. The all-zero list is treated separately and has generator 0. This is simply the defining property of a gcd, but it gives a useful physical reading: a proper block-darkness locus requires simultaneous cancellation of the entire finite marked future, not merely one selected residual or one mixed linear relation.

5 Fourier arithmetic for the marked baselines

The projector formula (3) reduces every local rank-one contraction to a finite root-of-unity sum. If

$$A = \alpha |x\rangle\langle y|,$$

then

$$P_\phi A P_\theta = \alpha (P_\phi |x\rangle)(\langle y| P_\theta), \tag{9}$$

and an ordered product of such rank-one factors contracts into a product of projector entries. In circulant coordinates these entries are exactly the marked sums $S_\theta(r)/n$. The Fourier algebra in (9) is routine; what matters operationally is that nonzero contributions must be added coherently before a zero test is made.

For prime powers, the local vanishing test can be stated compactly. Let $n = p^a$, $r \neq 0$, $d = \gcd(r, n)$ and $q = n/d = p^b$. If c_j counts the mask elements in residue class $j \pmod q$, then the standard cyclotomic divisibility argument gives

$$S_M(r) = 0 \iff c_j = c_{j+q/p} = \cdots = c_{j+(p-1)q/p} \quad (0 \le j < q/p). \tag{10}$$

This is a direct prime-power corollary of the classical root-of-unity machinery (Lam and Leung 2000). At prime order, a nonempty proper 0/1 mask cannot satisfy the corresponding equality at nonzero displacement. We use (10) only to certify local projector zeros in the examples; the physical amplitude functions remain separate data.

6 A rank-one reduction for a marked dark pair

We now specialise the marked model to one event. The scalar identity is a direct use of the standard rank-one inverse update; the point of the proposition is to record how its two amplitudes populate the block and arrow contents of the chosen marked realisation.

Proposition 6.1 (One-event reduction)**.** *Let H_0 be a Hermitian circulant and let distinct marked sites s, ℓ be exactly dark:*

$$\langle \ell | P_\theta | s \rangle = 0 \qquad \textit{for every spectral fibre } \theta.$$

Let $w_\theta = \langle s | P_\theta | s \rangle = \langle \ell | P_\theta | \ell \rangle > 0$ and define

$$q_\theta(z) = \frac{w_\theta}{z - \theta}, \qquad d(z) = \sum_\theta q_\theta(z).$$

On a one-parameter PID stratum let

$$A(t) = \big(a(t)|\ell\rangle + b(t)|s\rangle\big)\langle s|, \qquad (a, b) = A_\Sigma. \tag{11}$$

Use the regular fibre frames

$$x_\theta = P_\theta\big(a|\ell\rangle + b|s\rangle\big).$$

Then the declared one-event endpoint model has block readout and arrow coefficients

$$\Omega_\theta = a q_\theta, \qquad M_{\delta\gamma} = b q_\gamma, \tag{12}$$

and its grade-h block residual is

$$r_{h,\gamma} = a\, b^h d(z)^h q_\gamma(z). \tag{13}$$

Consequently

$$g_{\gamma,\Sigma} = \mathrm{norm}_\Sigma(a), \qquad h_{\delta\gamma,\Sigma} = \mathrm{norm}_\Sigma(b). \tag{14}$$

The marked scalar resolvent entry is

$$\langle \ell | (zI - H_0 - \varepsilon A)^{-1} | s \rangle = \frac{\varepsilon\, a(t) d(z)^2}{1 - \varepsilon\, b(t) d(z)}. \tag{15}$$

Proof. Circulancy makes every diagonal entry of P_θ equal to w_θ, while exact darkness gives $\langle \ell | P_\theta | s \rangle = 0$. Hence $P_\theta|s\rangle$ and $P_\theta|\ell\rangle$ are orthogonal and have the same nonzero norm. Unimodularity of (a, b) makes each displayed frame regular on the chosen stratum. Direct contraction gives (12), and induction on the event grade gives (13). The grade-zero coefficient has content (a), every later residual is divisible by a, and each declared arrow has content (b), proving (14).

For the scalar entry, write $G_0 = (zI - H_0)^{-1}$ and $v = a|\ell\rangle + b|s\rangle$, so $A = v\langle s|$. The Sherman–Morrison formula (Sherman and Morrison 1950; Hager 1989) gives

$$(zI - H_0 - \varepsilon v\langle s|)^{-1} = G_0 + \frac{\varepsilon G_0 v\langle s|G_0}{1 - \varepsilon\langle s|G_0 v}.$$

The baseline cross entries vanish, while $\langle s|G_0|s\rangle = \langle \ell|G_0|\ell\rangle = d(z)$. Therefore

$$\langle \ell|G_0 v = ad(z), \qquad \langle s|G_0 v = bd(z),$$

which yields (15). □

The scalar part of proposition 6.1 is therefore not a new rank-one theorem. Its use here is to expose a simple separation inside the declared model: the feed-forward amplitude a controls block darkness, while the return amplitude b controls the declared continuation graph.

7 Worked example on a published eight-site baseline

We use the eight-site oriented circulant with connection set

$$\mathcal{C} = \{1, 3\},$$

with Hermitian adjacency convention

$$(H_0)_{j,j+c} = \mathrm{i}, \qquad (H_0)_{j,j-c} = -\mathrm{i}, \qquad c \in \mathcal{C}.$$

The zero-transfer displacements 2 and 6 for this baseline appear in the public census of Song and Lin (2026, Appendix A); the baseline is not introduced here as a new graph. We mark $s = 0$ and $\ell = 2$.

The spectral fibres are

$$M_0 = \{0, 2, 4, 6\}, \qquad M_- = \{1, 3\}, \qquad M_+ = \{5, 7\}, \tag{16}$$

with poles $0, -2\sqrt{2}, +2\sqrt{2}$ and weights $1/2, 1/4, 1/4$. Each marked sum at displacement 2 vanishes. The common diagonal Green function is therefore

$$d(z) = \frac{1}{2z} + \frac{1}{4(z + 2\sqrt{2})} + \frac{1}{4(z - 2\sqrt{2})} = \frac{z^2 - 4}{z(z^2 - 8)}. \tag{17}$$

Consider the affine one-event family

$$A(t) = \big((1 - t)|2\rangle + (1 + t)|0\rangle\big)\langle 0|. \tag{18}$$

The amplitude pair is unimodular because $(1 - t) + (1 + t) = 2$. Hence the rank-one coefficient and all three projected image ranks remain one for every parameter value. By proposition 6.1,

$$g(t) = t - 1, \qquad h(t) = t + 1, \tag{19}$$

up to the chosen monic normalisation. Neither factor is an architecture factor.

The physical marked resolvent gives the same separation directly:

$$\langle 2|(zI - H_0 - \varepsilon A)^{-1}|0\rangle = \frac{\varepsilon(1 - t)d(z)^2}{1 - \varepsilon(1 + t)d(z)}. \tag{20}$$

At $t = 1$ the entire marked transfer is zero for every ε and z for which the resolvent exists. At $t = -1$ the denominator loses its return term, but the first response remains nonzero. Thus "dark block" and "no declared continuation" are genuinely different statements even in this simple family.

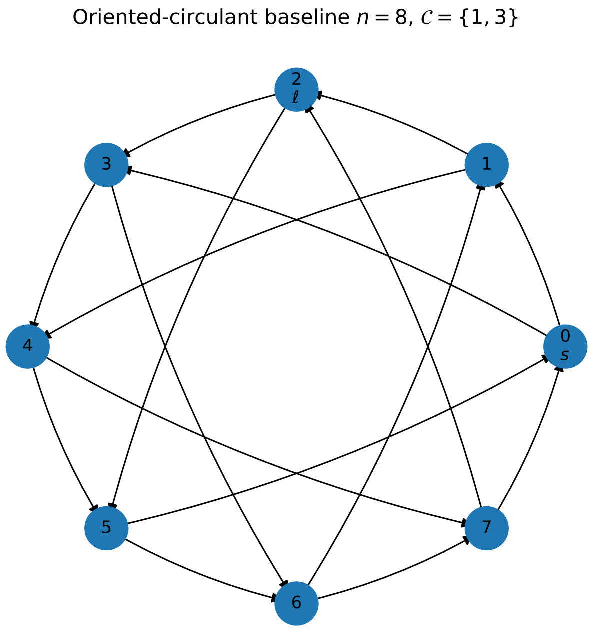

Figure 1: The oriented-circulant baseline $n = 8$, $\mathcal{C} = \{1, 3\}$, with marked source $s = 0$ and target $\ell = 2$. The zero-transfer status of displacement 2 is part of the public Song–Lin census.

Table 1: Specialisations of the affine family (18). All coefficient and projected-image ranks remain one.

Parameter	$K^{(0)}$	Nonzero declared arrows	$K_{\max}$
$t = 1$	all three blocks	all nine	all three blocks
$t = -1$	0	none	0
$t \neq \pm 1$	0	all nine	0

8 Static arithmetic does not determine the parameter loci

The preceding reduction also gives a useful boundary observation. Keep the same baseline, marked pair and declared model, but let $f(t) \in K[t]$ be arbitrary. If

$$A_f(t) = \big(f(t)|2\rangle + |0\rangle\big)\langle 0|,$$

then proposition 6.1 gives

$$g = \operatorname{norm}(f), \qquad h = 1.$$

If instead

$$\widetilde{A}_f(t) = \big(|2\rangle + f(t)|0\rangle\big)\langle 0|,$$

then

$$g = 1, \qquad h = \operatorname{norm}(f).$$

This is an immediate scaling consequence of the rank-one formula and content algebra, not a separate general impossibility theorem. Its role is conceptual: even complete static Fourier data for the baseline cannot determine the physical parameter loci when the permitted amplitude polynomial itself is allowed to vary. The baseline arithmetic certifies the dark geometry on which the perturbation acts; it does not replace the perturbation data.

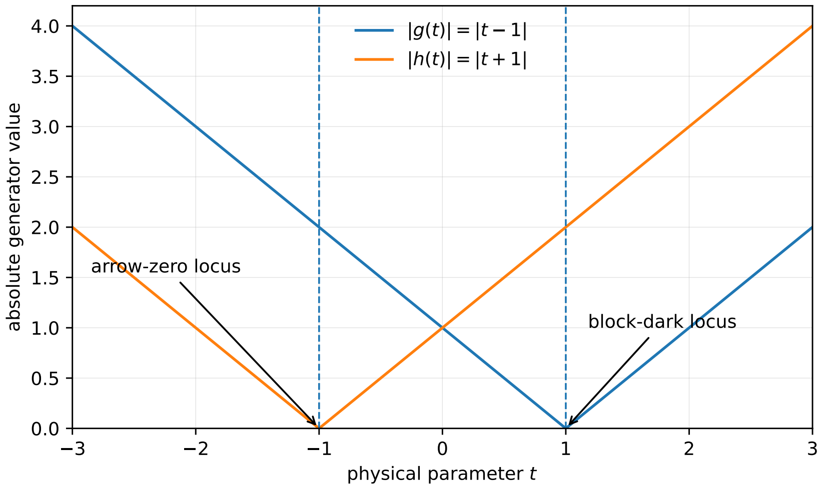

Figure 2: Illustration of the independent parameter loci in the affine family. The block generator $g = t - 1$ vanishes at $t = 1$, whereas the arrow generator $h = t + 1$ vanishes at $t = -1$. The plot is illustrative; the loci follow exactly from (19).

9 A bounded seven-block calculation on the eight-site baseline

We finally record an exact calculation for a fixed seven-block model on the same published eight-site baseline. We retain this bounded calculation in the main text because it is the paper's principal exact worked classification. We make no priority claim for its precise arithmetic pattern.

Consider

$$H(\varepsilon, t) = H_0 + \varepsilon A_1 + \varepsilon^2 A_2(t), \qquad A_1 = |1\rangle\langle 1| - |5\rangle\langle 5|, \qquad A_2(t) = t|x\rangle\langle y|. \tag{21}$$

On the nonzero-amplitude stratum $D(t)$ the declared endpoint/delay inventory contains seven reachable one-dimensional blocks:

Block	Meaning	Fibre	Degree
0	physical endpoint	0	1
1	physical endpoint	−	1
2	physical endpoint	+	1
3	unresolved delay memory	auxiliary	2
4	physical endpoint	0	2
5	physical endpoint	−	2
6	physical endpoint	+	2

At $t = 0$ the A_2 image vanishes and the canonical physical model must be rebuilt with only the three degree-one A_1 endpoint blocks; the four missing ambient coordinates are therefore not interpreted as new physical dark states.

The event locations are

$$(x, y) \in \{(1,4), (3,4), (5,4), (7,4), (6,1), (6,3), (6,5), (6,7)\},$$

and both marked directions $0 \to 2$ and $2 \to 0$ are retained. Let χ_4 denote the real primitive character modulo 4 on odd residues,

$$\chi_4(r) = \begin{cases} +1, & r \equiv 1 \pmod 4, \\ -1, & r \equiv 3 \pmod 4. \end{cases}$$

Table 2: Exact block-4 outcome in the fixed seven-block family on $D(t)$. The rule is marked ket/bra class plus χ_4 plus direction; it is not a χ_4-only statement.

Symbolic event location	Condition	Forward g_4, $0 \to 2$	Reverse g_4, $2 \to 0$
$x = r$ odd, $y = 4$	$\chi_4(r) = +1$, $r = 1, 5$	1	1
$x = r$ odd, $y = 4$	$\chi_4(r) = -1$, $r = 3, 7$	0	0
$x = 6$, $y = r$ odd	either value of $\chi_4(r)$	1	0

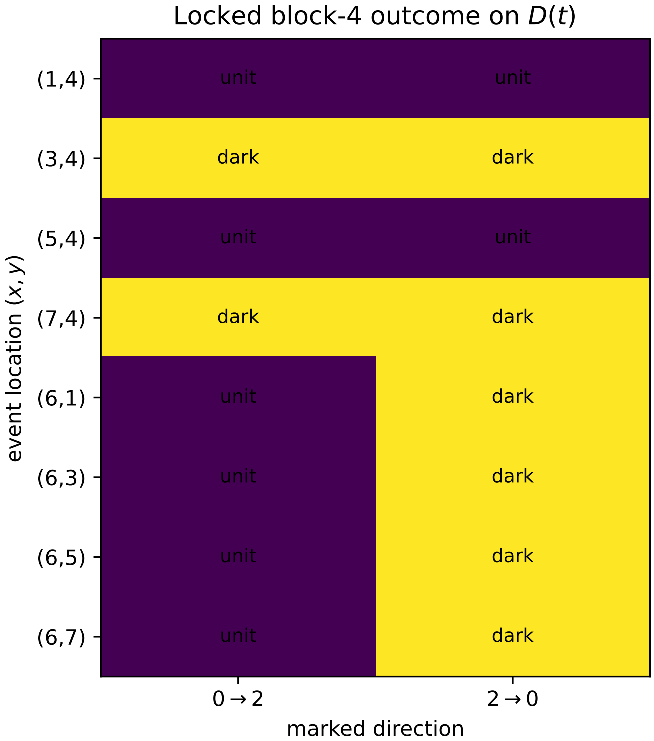

Figure 3: The sixteen marked components of the fixed seven-block calculation. “dark” means $g_4 = 0$ on $D(t)$; “unit” means $g_4 = 1$. The figure reports the bounded calculation only.

The exact calculation gives the block-4 table in table 2. Every other reachable block has unit generator on $D(t)$, and every declared arrow has unit generator there.

The arithmetic behind the exceptional block is elementary once the marked fibre and orientation are fixed. For the zero fibre $M_0 = \{0, 2, 4, 6\}$,

$$S_0(d) = \begin{cases} 4, & d \equiv 0 \pmod 4, \\ 0, & \text{otherwise}. \end{cases} \tag{22}$$

For odd r,

$$S_0(1 - r) = S_0(5 - r) = 2\big(1 + \chi_4(r)\big). \tag{23}$$

In the variable-ket family $(x, y) = (r, 4)$, the direct block-4 readouts and the A_2 contraction vanish. The outgoing A_1 vector is

$$A_1 G_0 P_0 |r\rangle = \frac{1 + \chi_4(r)}{4z} (|1\rangle - |5\rangle). \tag{24}$$

Hence $\chi_4(r) = -1$ gives a residual-zero dead end, while for $\chi_4(r) = +1$ the first future marked readout is nonzero. In the variable-bra family $(x, y) = (6, r)$, the block-4 endpoint is $P_0|6\rangle$, independent of the odd bra site. Its forward readout is $1/(2z)$, its reverse readout is zero, and all outgoing contractions vanish. This gives the reverse-only dark cases in table 2.

Across the sixteen components the exact ledger contains 112 block generators and 204 arrow generators. Before localisation the block generators are 48 units, 56 copies of the architecture factor t, and 8 zeros; the arrow generators are 140 units and 64 copies of t. On $D(t)$ the powers of t become units, leaving 104 unit block generators, 8 zero block-4 generators, and 204 unit arrows. In every zero case,

$$K^{(0)} = K_{\max} = \operatorname{span}\{e_4\};$$

in the remaining cases both spaces vanish. The mixed response factors $2t - \mathrm{i}$ and $2t + \mathrm{i}$ that occur in a larger global response calculation do not occur in any block or arrow generator of this model.

10 Discussion

The calculations above separate three levels of information that are easy to conflate.

First, zero transfer is a statement about a marked scalar channel on the baseline. For circulants it is efficiently certified by spectral fibres and root-of-unity sums. Secondly, after a perturbation model is declared, a particular coordinate block is dark only when its entire finite marked future vanishes. In a one-parameter PID model this is recorded by the content generator g_γ. Thirdly, structured survival imposes an additional requirement: every declared nonzero continuation of that dark block must remain inside the dark set. This is a common invariant-kernel condition, represented in one dimension by the forward closure of the nonzero-arrow graph.

None of these distinctions requires a new general observability or invariant-subspace theorem. Their value here is organisational. The graph-theoretic arithmetic determines exact local coefficients; standard finite observability packages the complete future; coefficient content exposes the physical parameter locus; and a standard common-invariant-kernel calculation enforces the declared structure.

The affine example makes another boundary explicit. The static zero-transfer data remain fixed while the parameter roots of g and h can be moved by changing the physical amplitude polynomial. A classification of static spectral masks alone therefore cannot determine parameterised darkness for a family in which those amplitudes are free data. Conversely, once the amplitudes and the declared model are fixed, the content calculation is exact and finite.

There are several limitations. The perturbation in (11) is directed and need not preserve Hermiticity or circulancy. The content-generator description is stated on one-dimensional PID strata; more general locally closed strata may require ideals rather than single polynomials. Higher-dimensional blocks replace coordinate contents by subspace/module questions. Multiple independently declared events allow more elaborate coherent cancellation, and arbitrary composite orders require a broader arithmetic analysis. These are outside the present scope.

11 Conclusion

Starting from known zero-transfer oriented-circulant baselines, we have organised a prescribed marked perturbation calculation around two questions: which individual declared blocks remain dark, and which of those dark blocks survive every declared continuation. Standard finite observation and coefficient content attach block and arrow parameter ideals to the model, while standard common invariant-kernel theory converts those data into structured survival. A one-event rank-one reduction shows transparently how feed-forward and return amplitudes occupy the two roles, and the published eight-site zero-transfer baseline provides a compact exact example in which their parameter loci are distinct. A separate bounded seven-block calculation illustrates how marked event type, Fourier arithmetic and direction combine in a fixed model.

The intended message is therefore not a new general observability theory. It is that zero-transfer arithmetic, parameter content and declared invariant structure can be assembled into a reproducible workflow for studying parameterised darkness on fixed graph baselines.

A Strata, architecture changes and generator normalisation

A strong architecture stratum fixes the ranks required by the declared construction, including projected coefficient-image ranks such as $\operatorname{rank}(P_\theta A_k)$ when those images define blocks. Exact rank conditions are locally closed, hence the coordinate ring takes the form $A_\Sigma = S^{-1}(R/I_\Sigma)$. A response zero inside such a stratum is distinct from a coefficient-rank collapse that changes the block inventory.

If $J \subseteq R$ is a block or arrow ideal, its stratum version is JA_Σ. In the one-parameter localisation $K[t, t^{-1}]$, powers of t are units and are removed by normalisation. The zero ideal is represented by generator 0; it is not assigned a monic generator. A nonzero ideal on a PID stratum is represented by a monic associate after removal of the specified stratum units.

For the locked family (21), $t \neq 0$ gives coefficient ranks

$$\operatorname{rank} A_1 = 2, \qquad \operatorname{rank}(P_\theta A_1) = 1, \qquad \operatorname{rank} A_2 = 1, \qquad \operatorname{rank}(P_\theta A_2) = 1$$

for each nonempty fibre. The physical endpoint/delay dimension is seven. At $t = 0$, $A_2 = 0$ and the physical model is rebuilt with only the three degree-one A_1 endpoint states. Algebraically specialising a fixed seven-coordinate ambient realisation to $t = 0$ is therefore not the same operation as rebuilding the physical structured model at that rank-collapse point.

B Content invariance under fixed coordinate changes

Let W be a finite-dimensional K-space of proper rational functions with a fixed parameter-independent denominator. Two fixed coordinate bases of W are related by some $C \in \mathrm{GL}_m(K)$. If a residual has coordinate vectors x and $y = Cx$, the entries of y lie in the ideal generated by the entries of x, while $x = C^{-1}y$ gives the reverse containment. Hence the coefficient ideal is independent of the chosen fixed rational-function basis.

Likewise, multiplication by a fixed monic polynomial $Q(z) \in K[z]$ preserves parameter content on its image: polynomial division recovers the original coefficient vector by a K-linear triangular map. This justifies clearing fixed baseline denominators. In contrast, clearing $1/z$ by the parameter-dependent denominator tz would replace the unit content (1) by the false factor (t), and is therefore inadmissible.

Coherent recombination must also precede content extraction. For example,

$$1 + (t - 1) = t$$

has content (t), whereas the sum of the separate ideals $(1)+(t-1)$ is the unit ideal. The shortest cancellation is $1 + (-1) = 0$ although both summand ideals are units. These elementary examples are why the paper keeps exact amplitudes until after declared contributions have been added.

C Order-nine cross-field example

For completeness, a second certified baseline uses $n = 9$ and connection set $\mathcal{C} = \{3\}$. The graph is the disjoint union of three directed triangles, so connectedness is not claimed. With marked sites $s = 0$, $\ell = 1$, the fibres are

$$\{0, 3, 6\}, \qquad \{1, 4, 7\}, \qquad \{2, 5, 8\},$$

at poles $0, -\sqrt{3}, +\sqrt{3}$. Each has weight $1/3$ and the source-target projector sums vanish. The diagonal Green function is

$$d(z) = \frac{z^2 - 1}{z(z^2 - 3)}.$$

Here the distinction between arithmetic and response fields is visible: the marked sums lie in $K_{\mathrm{Song}} = \mathbb{Q}(\zeta_9)$, whereas a convenient response field containing the real poles is $K = \mathbb{Q}(\zeta_{36})$. The same affine amplitudes $a = 1 - t$, $b = 1 + t$ give $g = t - 1$ and $h = t + 1$ by proposition 6.1.

D Exact verification summary

The source computation retained sixteen locked eight-site components, 112 block generators and 204 arrow generators. The paper package contains a compact data extract and an independent exact-check script. The script constructs the eight-site Hermitian adjacency, verifies

$$H_0^3 = 8H_0,$$

checks eigenvalue multiplicities $4, 2, 2$, verifies the $0 \to 2$ baseline moments through the finite horizon, reconstructs

$$d(z) = \frac{z^2 - 4}{z(z^2 - 8)},$$

and checks the first three event grades of (15) symbolically. It also verifies the locked generator counts and that precisely eight of the sixteen marked directions have dark block 4.

The exact ledger reports early unit-content witnesses for every non-dark block: 60 at grade 0, 32 at grade 1, 8 at grade 2 and 4 at grade 3. These grades are computational certificates in the fixed presentation, not invariants under arbitrary reparametrisation of the state or grade structure. The machine calculations support the finite census; they do not substitute for the all-parameter arguments in the main text.

Transparency statement

The mathematical organisation, manuscript drafting, and numerical diagnostic scripting in this document were developed with assistance from ChatGPT and Gemini. These tools are not authors and cannot take responsibility for correctness. The named human authors are responsible for checking the mathematics, validating computations, verifying citations, auditing generated code and data, and deciding whether the manuscript is suitable for dissemination or submission.

Data and code availability

The accompanying reproducibility package contains the figure-generation script, a compact JSON extract of the exact census used in the tables, and an independent symbolic verification script. The larger project ledgers from which the extract was prepared are not required to compile the manuscript. No numerical tolerance is used in the exact checks; floating-point evaluation is used only to draw the illustrative generator plot.

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